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3.2.6 Cost of Gram-Schmidt algorithms

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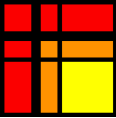
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Read Unit 3.2.6 of the notes. [\[LINK\]](#)

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Homework 3.2.6.1

1/1 point (graded)

Analyze the cost of the CGS algorithm in

$[A, R] := \text{GS}(A)$ (overwrites A with Q)

$A \rightarrow \left(\begin{array}{c|c} A_L & A_R \end{array} \right), R \rightarrow \left(\begin{array}{c|c} R_{TL} & R_{TR} \\ \hline 0 & R_{BR} \end{array} \right)$

A_L has 0 columns and R_{TL} is 0×0

while $n(A_L) < n(A)$

$\left(\begin{array}{c|c} A_L & A_R \end{array} \right) \rightarrow \left(\begin{array}{c|c} A_0 & a_1 \quad A_2 \end{array} \right), \left(\begin{array}{c|c} R_{TL} & R_{TR} \\ \hline 0 & R_{BR} \end{array} \right) \rightarrow \left(\begin{array}{c|c|c} R_{00} & r_{01} & R_{02} \\ \hline 0 & \rho_{11} & r_{12}^T \\ 0 & 0 & R_{22} \end{array} \right)$

CGS	MGS	MGS (alternative)
$r_{01} := A_0^H a_1$	$[a_1, r_{01}] = \text{Proj}_{\perp \text{to } \mathbf{Q}_{\text{MGS}}}(A_0, a_1)$	$\rho_{11} := \ a_1\ _2$
$a_1 := a_1 - A_0 r_{01}$	$\rho_{11} := \ a_1\ _2$	$a_1 := a_1 / \rho_{11}$
$\rho_{11} := \ a_1\ _2$	$q_1 := a_1 / \rho_{11}$	$r_{12}^T := a_1^H A_2$
$a_1 := a_1 / \rho_{11}$		$A_2 := A_2 - a_1 r_{12}^T$

$\left(\begin{array}{c|c} A_L & A_R \end{array} \right) \leftarrow \left(\begin{array}{c|c} A_0 & a_1 \quad A_2 \end{array} \right), \left(\begin{array}{c|c} R_{TL} & R_{TR} \\ \hline 0 & R_{BR} \end{array} \right) \leftarrow \left(\begin{array}{c|c|c} R_{00} & r_{01} & R_{02} \\ \hline 0 & \rho_{11} & r_{12}^T \\ 0 & 0 & R_{22} \end{array} \right)$

endwhile

(left-most algorithm) assuming that $A \in \mathbb{C}^{m \times n}$.

The cost of the CGS algorithm is, approximately,

☐ $2m^3$

☐ $2nm^2$

☐

MGS

☒

They require (approximately) the same computation



Solution:

For complete solution see [the notes](#).

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